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REVIEW ARTICLE

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Do Wechsler intelligence scales predict academic achievement in children with ADHD or autism? A systematic review and meta-analysis

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ABSTRACT

Intelligence tests predict academic achievement in typically developed children, however if this is the case also in children with attention-deficit/hyperactivity disorder (ADHD) and/or autism spectrum disorder (ASD) is not clear. This systematic review and meta-analysis examined if Wechsler intelligence scales predict academic achievement and/or grades in children, ages 6-16 years, with ADHD and/or ASD. We searched the databases PubMed, PsycINFO and Education Research Complete for studies published between 2000 and 2023. We used the Newcastle-Ottawa Scale to assess risk of bias. Narrative synthesis and meta-analysis were performed. Twelve studies (ADHD n = 1,834, ASD n = 176) were included in the review, and six samples (ADHD n = 1,112) of those were included in the meta-analyses. The results of the meta-analyses showed moderate overall weighted correlations between IQ and word reading, written language, and mathematics respectively. Similarly, the overall weighted correlations between processing speed and the aforementioned domains of academic achievement were moderate. Meta-analysis with additional Wechsler scales composite scores could not be conducted. In the narrative synthesis, Full Scale IQ was associated with academic achievement in both ADHD and ASD, and grades in ADHD. The limited number of ASD participants and the heterogeneity of the samples need to be considered when interpreting results. Generally, the results indicate that Wechsler scales are valuable in predicting academic achievement in children with ADHD or ASD. Motivation and other factors related with academic achievement need to be further explored in these groups.

KEYWORDS

Intelligence; academic achievement; Wechsler scales; ADHD; autism

Introduction

Rationale

Attention-deficit/hyperactivity disorder (ADHD) and autism spectrum disorder (ASD) are both considered neurodevelopmental disorders (NDDs), often identified and diagnosed in childhood, affecting the individual's development and behavior. Though more prevalent in males, ADHD and ASD affect both males and females, are heterogeneous in their clinical presentation and etiology, and frequently coexist (Leitner, 2014). Individual outcomes vary, but in many cases symptoms and impairment continue into adulthood (Thapar et al., 2017).

ADHD, which is the most common NDD, is characterized by inattention, hyperactivity, and impulsivity that interfere with a person's functioning (American Psychiatric Association, 2013). Findings of slightly lower mean FSIQ scores compared to controls have been reported in groups with ADHD (Agnew-Blais et al., 2020; Frazier et al., 2004), but others have suggested that lower FSIQ scores occur only in certain subgroups (Bridgett & Walker, 2006). FSIQ at an individual level in ADHD varies widely (Cadenas et al., 2020; Lindblad et al., 2013). Other cognitive deficits are related to executive function, including poor inhibitory control and poor working memory (Pievsky & McGrath, 2018; Ramos et al., 2020), and to processing speed (Cook et al., 2018). Furthermore, ADHD has been associated with unfavorable academic outcomes measured with achievement tests, grade point average, grade retention, school dropout rates, and years of education (Arnold et al., 2020; Fleming et al., 2017; Klein et al., 2012). Children and adolescents with ADHD display significantly lower academic achievement in mathematics, reading, and writing compared to

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their peers (Lawrence et al., 2021), and a systematic review has shown a negative association between ADHD and mathematical ability, stronger for inattentive symptoms (Tosto et al., 2015). Furthermore, better cognitive performance in memory, inhibitory control, and processing speed has been reported to be associated with better mathematic performance in children with ADHD (Kanevski et al., 2022). Working memory is consistently correlated with several aspects of literacy, whereas smaller correlations have been found between processing speed and literacy (McDougal et al., 2022).

Distinguished features of ASD are deficits in social communication and social interaction, combined with restrictive, repetitive behaviors and interests (American Psychiatric Association, 2013). ASD is, similar to ADHD, a highly heterogeneous disorder that occurs at all levels of intellectual functioning (Wolff et al., 2022). Intelligence and intellectual disability (ID) in ASD have been studied extensively. Initially considered highly prevalent in ASD, the rates of concurrent ID in ASD have during the last five decades diminished from about 70% to 30% (Wolff et al., 2022), and a recent meta-analysis reported rates of about 23% (Mutluer et al., 2022). Similar to intellectual functioning, function in other cognitive domains varies widely among individuals with ASD. Apart from social cognitive deficits (Braconnier & Siper, 2021), executive function deficits, including in flexibility and working memory (Lai et al., 2017), are common in ASD. Deficits in processing speed have also been reported (Mayes & Calhoun, 2008). Regular school environment poses challenges for children with ASD (Ashburner et al., 2010), and there is a need for educational interventions and accommodations (Hyman et al., 2020). Academic outcomes in ASD are variable and factors associated with favorable academic outcome are higher IQ, no working memory problems, less behavior problems, and better social skills (Keen et al., 2016).

The Wechsler Intelligence Scale for Children (WISC) is an individually administered standardized test to assess general intellectual functioning in children aged 6 to 16 years. The Wechsler scales are often the first choice by school psychologists and by clinical neuropsychologists, not only in North America and Europe, but globally (Oakland et al., 2016; Rabin et al., 2016). WISC is used for different purposes including for neuropsychological assessment in ADHD (Thaler et al., 2013) and ASD (Braconnier & Siper, 2021). The first edition was introduced in the 1940s (Wechsler, 1949) and updated versions have been published since, WISC-V (Wechsler, 2014) being the most recent. Besides scores of Verbal IQ (VIQ), Performance IQ (PIQ), and Full Scale IQ (FSIQ) featured in WISC, WISC-III (Wechsler, 1991) also included scores for four factor-analytically derived indexes called Verbal Comprehension Index (VCI), Perceptual Organization Index (POI), Freedom from Distractibility Index (FDI), and Processing Speed Index (PSI). FDI comprised the Arithmetic and Digit Span subtests, and was replaced in the subsequent WISC version, the WISC-IV (Wechsler, 2003a), with Working Memory Index (WMI), that comprised Digit Span and Letter-Number Sequencing to minimize the impact of mathematics (Kaufman et al., 2006). Similar to WISC-III, WISC-IV has also a four-factor structure reflected in the indexes VCI, Perceptual Reasoning Index (PRI), WMI, and PSI. In both WISC-III and WISC-IV, PSI is based on the Coding and the Symbol Search subtests, while WMI in WISC-V comprises Digit Span and a new, visual, working memory subtest (Picture Span). The correlation between FSIQ in WISC-III and in WISC-IV is strong (Wechsler, 2003b). The latest edition of WISC, the WISC-V (Wechsler, 2014), has a five factor-structure including the indexes VCI, Visual Spatial Index, Fluid Reasoning Index, WMI and PSI. Adolescents from 16 years and adults can be assessed with the Wechsler Adult Intelligence Scale (WAIS; Wechsler, 2008). In addition to the aforementioned tests, the Wechsler Abbreviated Scale of Intelligence (WASI; Wechsler, 1999) and its subsequent version WASI-II (Wechsler, 2011) are both abbreviated measures for estimation of intellectual functioning to be used when comprehensive assessment is not possible or needed, or for screening purposes to determine if further assessment is required.

Intelligence (defined as IQ) is a robust predictor of academic achievement in the general population (Deary et al., 2007; Neisser et al., 1996; Plomin & von Stumm, 2018). Intelligence tests predict academic achievement (Deary et al., 2007) and school grades (Roth et al., 2015). Furthermore, contemporary research has proposed that the relationship between intelligence and academic achievement might be bidirectional rather than unidirectional, i.e., that intelligence and achievement influence each other during development. This also applies to working memory and academic achievement. These relationships seem stronger during certain developmental periods, and schooling, particularly in the primary and secondary years, may facilitate this bidirectionality (Peng & Kievit, 2020). However, these relationships might be weaker or absent among children with learning disabilities (Ferrer et al., 2010; Quinn et al., 2020).

Studies examining the association between WISC- or WASI-results and academic achievement in ADHD and ASD, point out intelligence as a predictor of academic achievement (Alloway & Stein, 2014; Mayes & Calhoun, 2008). In clinical settings, there is a notion that children with NDD and normal or higher IQ would do better in school than children with NDD and lower IQ, which is plausible and maybe true for some individuals. However, despite correlations between IQ and academic achievement in ASD (Mayes & Calhoun, 2008) and ADHD (Mayes & Calhoun, 2007), discrepancy between IQ and academic achievement is relatively common in ASD (Jones et al., 2009; Kim et al., 2018; Mayes et al., 2020), and ADHD (Mayes et al., 2020), encompassing mostly underachievement. Furthermore, IQ and adaptive functioning are positively correlated in the general population, but this relationship is less clear in children and adolescents with IQ $>$ 70 and ADHD, ASD or other NDDs, as higher IQ in these groups is not consistently associated with improved adaptive functioning (Alvares et al., 2020; Åsberg Johnels et al., 2021). These findings together highlight the need for clarifying the role of IQ in academic achievement. A systematic review and meta-analysis on this topic could add to the knowledge on how FSIQ and other results from the

Wechsler scales are related to academic achievement in children with ADHD and/or ASD.

Objectives

The objective of this systematic review and meta-analysis is to examine if Wechsler intelligence scales predict academic achievement and/or grades in children with ADHD and/or ASD.

Methods

Protocol and registration

The systematic review has been registered in PROSPERO (International prospective register of systematic reviews, registration number CRD42022338617). The PRISMA (the Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines (Page et al., 2021) were followed in this review.

Eligibility criteria

Studies were considered eligible if they were observational (case-control, cross-sectional, longitudinal), included original data, were published in peer-reviewed publications between 01-01-2000 and 24-02-2023, and were written in English. The eligible participants were children attending school, clinically assessed at ages 6-16 years and diagnosed with ADHD or/and ASD. Studies including children aged 0-5 years, adolescents from 17 years and older, and adults were considered ineligible. Furthermore, studies were eligible if they reported quantitative measures of intellectual functioning; more specifically Wechsler scales (WISC, WASI). Academic achievement and/or grades were the outcomes of interest, and, thus, studies reporting quantitative measures of those were considered eligible. Studies on children with intellectual disability (ID) alone or as comorbidity, and studies where data from children with ID could not be excluded from correlation and/or regression analyses were also considered ineligible. If ADHD and/or ASD diagnosis was based only on parental ratings of symptoms, the studies were considered ineligible.

Information sources

PubMed, PsycINFO and Education Research Complete were searched for studies.

Search strategy

The search was first conducted on the 14th of June 2022 and updated on the 24th of February 2023. The search strategy included a combination of 3 key elements: i) intellectual functioning or Wechsler scales, ii) academic achievement, and iii) diagnosis (ADHD, ASD). The search terms for each database are presented in supplementary Table S1.

Study selection

Two reviewers (MM, EB) independently screened the retrieved studies, based on their title and abstract to determine if inclusion or exclusion criteria were met. The reviewers were blind to each other's decisions during this process and resolved disagreements afterwards. Full-text was retrieved for studies identified as potentially eligible, and for studies where decision was not possible based on title and/or abstract. The two reviewers (MM, EB) independently reviewed all full-text for eligibility and resolved disagreements by discussion. Figure 1 presents the study selection process.

Data collection process

The following information was extracted: general information about the study (author, year, country), study design, sample size, participant demographics (sex, age at test, age at assessment of academic achievement/grade), diagnosis (ADHD and/or ASD), comorbidity, test, test result, measure of academic achievement/grades, outcome (result of academic achievement/grades). Data were extracted by two reviewers (MM, EB).

Risk of bias assessment

Risk of bias assessment was conducted independently by two reviewers (MM, EB) and any disagreements were resolved by discussion. The Newcastle-Ottawa Scale (NOS; Wells et al., 2009), that is designed for assessing the quality of non-randomized studies, was used in this review. NOS has been developed for assessing case-control and cohort studies but has also been adapted for assessing the quality of cross-sectional studies. Since the objective of the review was to study association between WISC/WASI scores and achievement, comparisons of test scores between ADHD, ASD, and other clinical groups or typically developing controls were not included in the review. For this reason, case-control studies were evaluated as cross-sectional in the risk of bias assessment, if they did not explicitly compare the strength of correlations or associations between groups. We used the NOS adaptation for cross-sectional studies provided by Stewart et al. (2017), assessing the included studies based on the selection of study groups, comparability of the groups, and outcome. We tailored the questions in the comparability section to include school support and socioeconomic status as confounding factors, since they are relevant for the present review. In accordance with (Kanevski et al., 2022), a score of $\geq$5 to 9 stars/points indicated low risk of bias, 3-4 stars medium risk of bias, and $<$3 stars high risk of bias.

Synthesis methods

The extracted data were tabulated in two tables, one for ADHD studies and one for ASD studies. Data from other diagnostic groups or typically developing controls in the studies were not included in the synthesis. A narrative synthesis of the included studies was conducted. We decided a priori that meta-analysis examining correlations between

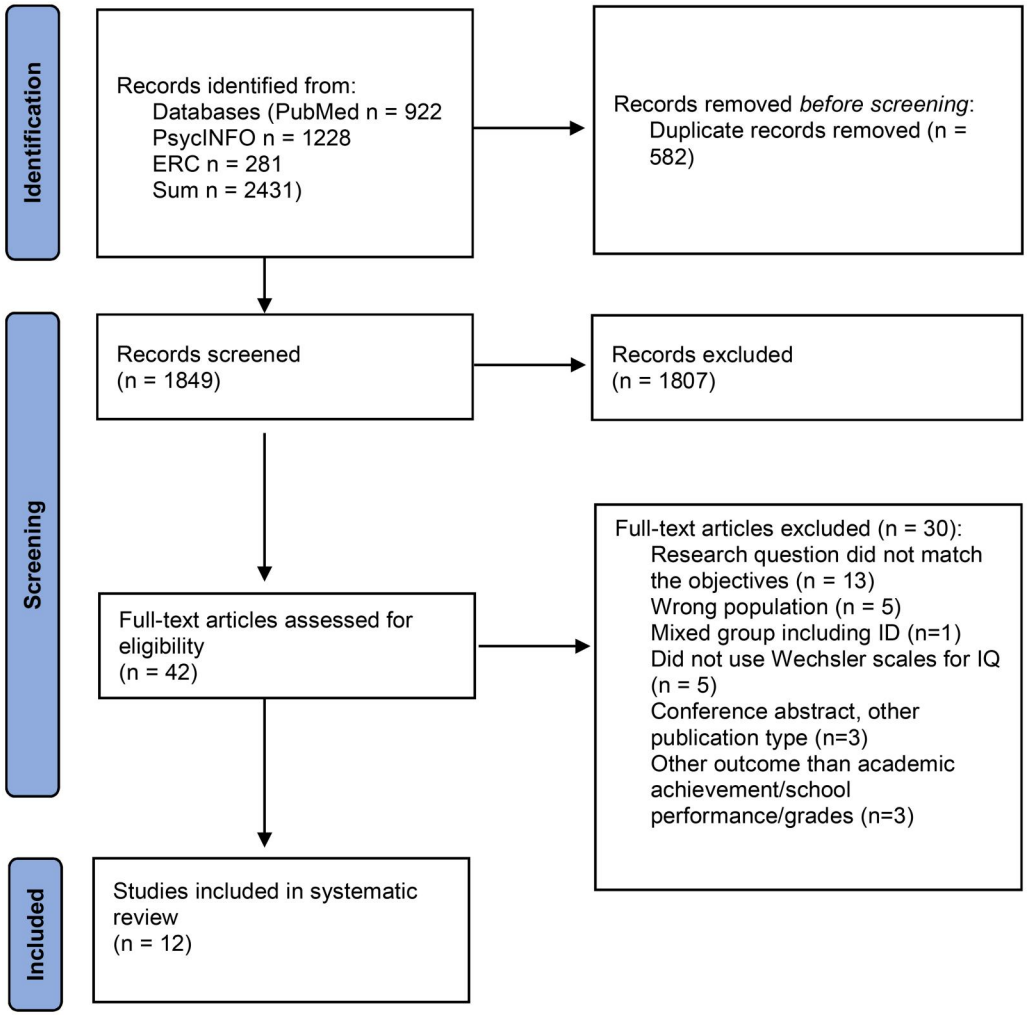


Figure 1. PRISMA flowchart, systematic review steps.

FSIQ (or other Wechsler scales indexes) and academic achievement or grades required at least four studies or samples to be performed. The meta-analyses were conducted with the tool Meta-Essentials (Suurmond et al., 2017), using a random effects model due to expected heterogeneity across studies. The I^2 statistic was used to assess heterogeneity. Publication bias was assessed with funnel plots including the results of a trim-and-fill procedure (Duval & Tweedie, 2000a; 2000b) provided by the Meta-Essentials software.

Results

Study selection

Details of the screening and exclusion processes are presented in the PRISMA flowchart (Figure 1). The search rendered 2,431 results. After de-duplication, titles and abstracts of 1,849 records were screened by MM and EB, and 1,807 records were excluded. This resulted in 42 studies for full text screening. Of those, 12 were included in the systematic review.

Study characteristics

Table 1 and 2 present the main characteristics of the included studies.

Three studies (Assouline et al., 2012; Mayes & Calhoun, 2003; 2008) included samples with ASD with a total of 176 participants. The rest of the studies (Alloway & Stein, 2014; Calub et al., 2019; Carrasco et al., 2022; Ek et al., 2011; Krane & Tannock, 2001; Mayes & Calhoun, 2007; Tallberg et al., 2021; Thaler et al., 2013; Zendarski et al., 2017) included samples with ADHD with a total of 1,834 participants. Zendarski et al. (2017) reports ASD as comorbidity in 33% of the ADHD sample, but results for this group are not separated from subjects with ADHD without ASD. Alloway and Stein (2014), Carrasco et al. (2022), Krane and Tannock (2001), and Tallberg et al. (2021) reported that ADHD medication in the samples was discontinued 24h prior to assessment in, while Mayes and Calhoun (2007) reported that participants did not take their morning dose of ADHD medication before assessment.

Several versions of WISC were used in the studies, except for two studies (Alloway & Stein, 2014; Zendarski et al., 2017) that used WASI. As regards outcomes, two studies used school grades; overall after grade 9 (Ek et al., 2011) or school grades 3-4 years after initial cognitive assessment (Tallberg et al., 2021). The rest of the studies used instruments to assess academic achievement. The Wechsler Individual Achievement Test (WIAT; Wechsler, 1992, 2002, 2009) was used as outcome

Table 1. Attention-Deficit/Hyperactivity Disorder (ADHD) – Study characteristics and main results.

Author Year Country Study design	N (Males) Age years Mean (SD) **ADHD subtype**	IQ (Mean, SD)	Measures academic achievement	Result	Risk of bias
Alloway and Stein (2014) USA, UK Case-control	83 (71) 9.9 (11.98) ADHD-C 100%	WASI Verbal IQ - Vocabulary (38.52, 10.60)a Non-verbal IQ: Matrix (41.00, 9.52)[a]	WIAT-II Spelling and arithmetic test	*Correlations* Vocabulary Arithmetic .59; Spelling .63 Matrix Arithmetic.49; Spelling .45 *Regression* Vocabulary accounted for 38.3% of the variance in spelling and 34.2% of the variance in arithmetic.	Medium
Calub et al. (2019) USA Case-control	28 (28) 9.54 (1.26) ADHD-C 100%	WISC-IV FSIQ (105.21, 10.33) VCI (108.10, 10.02) PRI (106.61, 13.10) WMI (100.64, 10.43) PSI (96.64, 10.73)	KTEA-I-NU/KTEA-II Reading and Math Composite scores	The combined effects of WMI and VCI accounted for 50% of the ADHD/Academic achievement relation.	Low
Carrasco et al. (2022) USA New Zealand Cross-sectional	The study included 4 samples, here numbered i-iv. Total N (males) 434 (322) Mean age 100.19 – 109.72 months ADHD-I 20.5% –59.5% ADHD-H 2.7%- 5.7% ADHD-C 37.8% –73.9%	i WISC-III, ii-iv WISC-IV Mean FSIQ 90.48-99.04 Mean Estimated IQ in sample i 90.36 Mean estimated IQ in samples ii-iv 93.34-99.04 Mean PSI 88.68 – 98.42	i WRAT-3 (Tan) ii WIAT-II iii WRAT-4(Green) iv WIAT-III Word reading, Spelling, Math computation	*Correlations* Estimated IQ and word reading 0.27-0.38 Estimated IQ and spelling 0.20-0.36 Estimated IQ and math computation 0.29-0.37 PSI and word Reading 0.17-0.26 PSI and spelling 0.23-0.31 PSI and math computation 0.29-0.36 *Regression* Estimated IQ explained significant variance in academic achievement, except for spelling in one sample.	Medium
Ek et al. (2011) Sweden Longitudinal	42 children WISC (grade available for 39 children) IQ test: 10-11 Grades: 16 ADHD Subtypes NR Comorbidity NR	WISC-III FSIQ (86.5, 16.5) VIQ (89.4, 14.4) PIQ (86.2, 19.3) VCI (91.5, 14.9) POI (89.7, 18.1) FDI (84.3, 14.7) PSI (84.3, 16.2)	Final grade sum after 9 years of compulsory school	*Correlations with overall grade* FSIQ .37 VIQ .36 PIQ .23 VCI .32 POI .23 FFD .28 PSI .45	Low
Krane and Tannock (2001) Canada Case-control	188 (158) 8.7 (1.72) ADHD-I 29.8% ADHD-H 20.2% ADHD-C 50.0%	WISC-III FSIQ (98.99, 13.59) VIQ (99.54, 13.85) PIQ (98.39, 15.11) VCI (100.12, 14.27) POI (99.64, 14.86) PSI (97.56, 15.92) FDI (95.70, 12.53)	WRAT-3 Reading and Arithmetic subtests WRMT-R Word Attack and Word Identification	*Correlations* FDI Reading Compositeb .38 WRAT Arithmetic .55	Medium
Mayes and Calhoun (2007) USA Cross-sectional	678 (74% in WISC-III sample, 64% in WISC-IV sample) 9 (2.0-2.5) ADHD-C n = 514 ADHD-I n = 164	WISC-III sample (n = 586), Mean FSIQ 104 WISC-IV sample (n = 92), Mean FSIQ 108	WIAT or WIAT-II Basic or Word Reading, Reading Comprehension, Numerical Operations, Written Expression	*Correlations* FSIQ and achievement scores $r = 0.47$- 0.65 (WISC-III), $r = 0.54$- 0.76 (WISC-IV). VCI and achievement scores $r = 0.44 – 0.64$ (WISC-III), $r = 0.43$-0.64 (WISC-IV). POI/PRI and achievement scores $r = 0.28 – 0.45$ (WISC-III), $r = 0.38$-0.62 (WISC-IV). FDI/WMI and achievement scores $r = 0.42 – 0.62$ (WISC-III), $r = 0.53$ to 0.64 (WISC-IV).	Medium

(*continued*)

Table 1. Continued.

Author Year Country Study design	N (Males) Age years Mean (SD) **ADHD subtype**	IQ (Mean, SD)	Measures academic achievement	Result	Risk of bias
				PSI and achievement scores $r = 0.34 - 0.43$ (WISC-III), $r = 0.33$ to 0.52 (WISC-IV). FSIQ was the best overall single predictor of achievement, accounting for 22% to 58% of the variance in reading, math, and writing. FDI/WMI and VCI scores were more correlated with achievement than POI/PRI and PSI. In stepwise regression analysis with FSIQ, all index and subtest scores as predictors, POI/PRI index and subtest scores did not significantly predict achievement.	
Tallberg et al. (2021) Sweden Longitudinal	125 (70%) 11.40 (3.27) ADHD subtypes NR	WISC-IV (n = 120) or WAIS-IV (n = 5) FSIQ (88.9, 12.67) VCI (93.43, 13.36) PRI (97.06, 14.04) WMI (82.03, 12.95) PSI (88.69, 13.59)	Grades	VCI at baseline was a significant predictor of having a majority of passing grades at follow-up. Higher VCI at follow-up was associated with higher likelihood of receiving a simple majority of passing grades at follow-up.	Medium
Thaler et al. (2013) USA Cross-sectional	189 (72.5%) 9.9 (2.8) ADHD-I 56.1% ADHD-C 43.9%	WISC-IV (N = 189) FSIQ (101.8, 12.0) VCI (104.7, 14.1) PRI (103.9, 12.0) WMI (97.0, 14.1) PSI (95.7, 13.9) Clusters (n = 129 with WJ-III ACH data) Reduced PSI (n = 57), High Average (n = 25), Below Average WMI/PSI (n = 19), Superior VCI (n = 14), Average PSI (n = 14)	WJ-III ACH available for 129 children	*One-way ANOVAs of subtests* Reading fluency: Reduced PSI and Below Average WMI/PSI clusters performed poorer than High Average and Superior VCI clusters. Math fluency: Reduced PSI and Below Average WMI/PSI clusters performed poorer than the High Average cluster. Applied problems: The Below Average WMI/PSI cluster performed poorer than the Reduced PSI and Superior VCI clusters.	Medium
Zendarski et al. (2017) Australia Cross-sectional	130 (89%) 13.7 (1.1) ADHD subtypes NR	WASI-II IQ M = 100.4 (15.8) WISC-IV backward Digit span subtest was used to measure working memory	NAPLAN (reading, writing, spelling, grammar and punctuation, numeracy) WRAT-4 The 7-item Academic Competence Scale	IQ explained the greatest proportion of variance in NAPLAN scores across all domains. Strongest effects: grammar and punctuation scores (standardized MD according to one SD higher IQ [b] .68; $p < .01$). Weakest effects: persuasive writing (β .31; $p = .04$) and spelling (β .31; $p = .04$). Higher IQ was associated with higher numeracy (β .62; $p < .01$), and higher reading (β .64; $p < .01$) scores. Working memory was not independently related to NAPLAN achievement.	Low

ADHD-C = Combined subtype; ADHD-H = Hyperactive-impulsive subtype; ADHD-I = Inattentive subtype; ANOVA = Analysis of Variance; ASD = Autism Spectrum Disorder; FDI = Freedom From Distractibility Index; FSIQ = Full Scale IQ; KTEA = Kaufman Test of Educational Achievement; NAPLAN = The National Assessment Program – Literacy and Numeracy; NR = Not reported; PIQ = Performance IQ; POI = Perceptual Organization Index; PRI = Perceptual Reasoning Index; PSI = Processing Speed Index; VCI = Verbal Comprehension Index; VIQ = Verbal IQ; WASI = Wechsler Abbreviated Scale of Intelligence; WISC = Wechsler Intelligence Scale for Children; WJ-III ACH = Woodcock-Johnson III Tests of Achievement; WMI = Working Memory Index; WIAT = Wechsler Individual Achievement Test; WRAT = Wide Range Achievement Test; WRMT-R = Woodcock Reading Mastery Tests -Revised.

[a]T-scores (M = 50, SD = 10).

[b]Comprised of the average score from WRAT-3 Reading, WRMT Word Attack and Word Identification subtests.

Table 2. Autism Spectrum Disorder (ASD) - Study characteristics and main results.

Author Year Country Study design	N (Males) Age years Mean (SD)	IQ (Mean, SD)	Measures academic achievement	Result	Risk of bias
Assouline et al. (2012) USA Cross-sectional	59 (50) (autism 24; AS 23; PDD-NOS 12) 10:7 (3.19)	WISC-IV (n = 48) VCI (125.1, 15.1) PRI (123.7, 13.3) WMI (111.3, 13.7) PSI (96.6, 15.7)	WJ-III ACH Broad reading, Broad math, Broad written language	*Correlations* VCI: Written language −0.034; reading 0.224; math 0.02 PRI: Written language −0.209; reading 0.239; math 0.333 WMI: Written language −0.496; reading 0.401; math 0.495 PSI: Written language −0.374; reading 0.713; math 0.475 *Regression** WMI and PSI accounted for 61% of the variance in reading. WMI predicted 25% of the variance in written language.	Medium
Mayes and Calhoun (2003) USA Cross-sectional	63 (NR) 8.5	WISC-III 33% IQ < 80 66% IQ ≥ 80	WIAT Basic Reading, Reading Comprehension, Numerical Operations, Spelling, Written Expression	Low IQ group (IQ < 80): Reading decoding exceeded IQ ($t = 3.7$, $p < .05$), but math and spelling did not. High IQ group (IQ ≥ 80): Written Expression was lower than IQ ($t = 4.9$, $p < .05$). Scores in the other subtests were consistent with IQ.	Medium
Mayes and Calhoun (2008) USA Cross-sectional	54 (48) 8.2 (2.0)	WISC-IV FSIQ (101,19) GAI (113,19) VCI (107,19) PRI (115,16) WMI (89,15) PSI (85,18)	WIAT-II Word Reading, Reading Comprehension, Numerical Operations, Written Expression	*Correlations* FSIQ Word Reading 0.64 Reading Comprehension 0.68 Numerical Operations 0.80 Written Expression 0.75 FSIQ – achievement $r = .64$-$.80$. GAI – achievement $r = .60$-$.78$. FSIQ was the best overall single predictor of reading, math and writing. Highest index-achievement correlations: WMI for reading, PRI for math, and PSI for written expression. Differences were, however, small.	Medium

AS = Asperger syndrome; FSIQ = Full Scale IQ; GAI = General Ability Index; NR = Not reported; PDD-NOS = Pervasive Developmental Disorder – Not Otherwise Specified; PRI = Perceptual Reasoning Index; PSI = Processing Speed Index; VCI = Verbal Comprehension Index; WIAT = Wechsler Individual Achievement Test; WISC = Wechsler Intelligence Scale for Children; WJ-III ACH = Woodcock-Johnson III Tests of Achievement; WMI = Working Memory Index.
*Subsample with WAIS-III included in the regression.

measure in several studies (Alloway & Stein, 2014; Mayes & Calhoun, 2003, 2007; 2008), while the Wide Range Achievement Test (WRAT; Jastak & Wilkinson, 1984; Wilkinson, 1993; Wilkinson & Robertson, 2006) was used in others (Krane & Tannock, 2001; Zendarski et al., 2017). Carrasco et al. (2022) included four different samples, using WRAT in two of those and WIAT in the other two. Woodcock-Johnson III Tests of Achievement (WJ-III ACH; Woodcock et al., 2001) was used in two studies (Assouline et al., 2012; Thaler et al., 2013). In addition, the Kaufman Test of Educational Achievement (KTEA; Kaufman & Kaufman, 2004; Kaufman & Kaufman, 1998) was used in Calub et al. (2019), Woodcock Reading Mastery Tests –Revised (WRMT-R; Woodcock, 1987) in Krane and Tannock (2001), the National Assessment Program – Literacy and Numeracy (NAPLAN; Australian Curriculum Assessment & Reporting Authority, 2015), and the 7-item Academic Competence Scale (Elliott & Gresham, 2013) in Zendarski et al. (2017).

Risk of bias

NOS results of risk of bias assessment for each study are presented in Table 1 and 2. Three of the studies had low risk of bias and nine studies had medium risk of bias. In general, the assessment of outcome in the studies was of high quality and the included samples were considered representative for the children with the disorder of interest. However, controlling for confounding factors was often missed.

Narrative synthesis of included studies

Overall findings of ADHD studies

In all, the included ADHD studies highlight FSIQ/IQ as a predictor of academic achievement (Carrasco et al., 2022; Ek et al., 2011; Mayes & Calhoun, 2007; Zendarski et al., 2017). Working memory (Calub et al., 2019; Carrasco et al., 2022; Krane & Tannock, 2001; Mayes & Calhoun, 2007), vocabulary, verbal comprehension and verbal reasoning (Calub et al., 2019; Carrasco et al., 2022; Mayes & Calhoun, 2007) are implicated in academic achievement and emerge as predictors, but one study did not find correlations between working memory and achievement (Zendarski et al., 2017). The impact of processing speed is less clear, and this ability might contribute to achievement to a lower degree than other abilities (Carrasco et al., 2022; Mayes & Calhoun, 2007). However, one study showed moderate correlations

between processing speed and overall grade (Ek et al., 2011), and reduced processing speed has been suggested to affect reading and math fluency (Thaler et al., 2013). Longitudinal studies examining WISC-results and academic achievement and grades are few (Ek et al., 2011; Tallberg et al., 2021).

Specific results of ADHD studies

As regards the findings of the included studies, Calub et al. (2019) found that WMI and VCI (WISC-IV) combined accounted for 50% of the relation between ADHD and academic achievement, of which 30% was contribution of WMI and 20% of VCI. Also, Mayes and Calhoun (2007) focused on academic achievement in a large sample of children with IQ >80. They reported significant correlations between FSIQ and all WIAT scales ($p = .0001$) for both the WISC–III and WISC–IV. FSIQ accounted for 22% to 58% of the variance in achievement scores and was the best overall single predictor of achievement. Entering the four index scores in a multivariate analysis the explained variance (30% – 59%) was somewhat higher than for FSIQ alone. FDI/WMI and VCI were more correlated with reading, math, and written expression than POI/PRI and PSI. Furthermore, Alloway and Stein (2014) found that WASI Vocabulary and Matrix correlated significantly with arithmetic and spelling from WIAT-II. In the regression analysis, Vocabulary accounted for 38.3% of the variance in spelling and 34.2% of the variance in arithmetic. Carrasco et al. (2022) found significant correlations between estimated IQ and word reading, spelling, and math computation domains in four samples with ADHD. The results of regression analyses showed that estimated IQ was a significant predictor of variance in academic achievement, except for spelling in the sample with larger proportions of boys and ADHD combined type. Semantic language, measured with the WISC subtest Vocabulary, was a predictor of word reading across samples, and predicted spelling in three samples, and math computation in one. Age and verbal working memory were also predictors of academic achievement, while the influence of the severity of ADHD symptoms on academic achievement was limited. Likewise, the contribution of processing speed was restricted. Explanatory hierarchical regression analyses showed that processing speed only explained additional variance in spelling in two samples and math computation in one sample. It did not explain additional variance in word reading. Moreover, Zendarski et al. (2017) showed that IQ explained the greatest proportion of variance in NAPLAN scores with strongest effects for grammar and punctuation scores and weakest effects for persuasive writing and spelling. Higher IQ was associated with higher numeracy and reading scores. Working memory measured with WISC-IV Digit Span was not independently related to any of NAPLAN domains. In addition, Krane and Tannock (2001) reported correlations between WISC-III FDI and a reading composite comprised of the average score from WRAT-3 Reading, WRMT-R Word Attack and WRMT-R Word Identification subtest, and between FDI and WRAT Arithmetic. Last, Thaler et al. (2013) found a five-cluster solution through a hierarchical cluster analysis. The clusters "Reduced Processing Speed Index" and "Below Average Working Memory Index/Processing Speed Index" performed poorer than the clusters "High Average" and "Superior Verbal Comprehension Index" on the reading fluency. On math fluency, they found that the cluster "Reduced Processing Speed Index" and the cluster "Below Average Working Memory Index/Processing Speed Index" also performed poorer than did the "High Average" cluster.

As regards studies using grades as outcome, Ek et al. (2011) and Tallberg et al. (2021) examined the association between WISC-results and grades. Ek et al. (2011) found significant correlations between FSIQ, VIQ, VCI at school grade 4 and overall grade at school grade 9. Tallberg et al. (2021) included WISC-indexes and found that baseline VCI significantly predicted having a majority of passing grades at follow-up. The explanatory effect of the model was 36%. Furthermore, they also found an association between VCI scores obtained at follow-up and the likelihood of passing grades at follow-up.

Overall findings of ASD studies

In all, the number of studies examining the relationship between WISC scores and academic achievement in ASD is limited, and none of those has used grades as achievement outcome and none included a longitudinal design. FSIQ (Mayes & Calhoun, 2008), WMI, and PSI (Assouline et al., 2012; Mayes & Calhoun, 2008) emerge as significant concurrent predictors of academic achievement in ASD, while VCI and PRI (Assouline et al., 2012; Mayes & Calhoun, 2008) are also implicated in achievement but the limited number of studies does not allow conclusions to be drawn on their contribution.

Specific results of ASD studies

As regards the findings of the included studies, Assouline et al. (2012) reported in a group of intellectually gifted students with ASD that WISC-IV WMI and PSI correlated with written language, reading, and mathematics assessed with WJ-III ACH. In contrast, PRI only correlated with oral language, while VCI did not correlate with any academic achievement scores. In the regression analyses, WMI and PSI explained 61% of the variance in reading, while WMI predicted 25% of the variance in written language. In addition, PSI, fine-motor coordination, and participation in talented and gifted programs together predicted 60% of the variance in mathematics. PRI predicted 22% of the variance in oral language. Additionally, Mayes and Calhoun (2008) have reported correlations between WISC-IV FSIQ and WIAT-II achievement ($r = 0.64 - 0.80$) accounting for 41 - 64% of the variance. FSIQ was the best overall single predictor of reading, math, and writing achievement. On index level, they found an association between PRI and achievement, similar with FSIQ, in the younger cohort (age 6–7 years). However, for the older half of the study group PRI had the lowest overall correlation with achievement, and FSIQ the highest. Finally, dividing a sample in a subgroup with IQ <80 and a subgroup with IQ ≥80 (measured with

WISC-III), Mayes and Calhoun (2003) found that, in the IQ < 80 group, WIAT reading decoding significantly exceeded IQ, whereas math and spelling scores were in line with IQ. For children in the IQ ≥ 80 group, achievement scores did not differ from IQ, except for written expression scores that were significantly lower than IQ, and lower than reading decoding, reading comprehension, math, and spelling scores.

Meta-analysis

Meta-analysis was possible to conduct in ADHD samples for FSIQ and academic achievement, and for PSI and academic achievement. FSIQ, estimated IQ, and PSI from WISC-III and WISC-IV were used in the analysis. Corrected correlation coefficients for WISC-III FSIQ and WISC-IV FSIQ (r = .89), and for WISC-III PSI and WISC-IV PSI (r = .81) are high (Wechsler, 2003b), enabling these measures to be used in the meta-analysis to measure FSIQ and processing speed, respectively. Academic achievement in the included samples was measured with editions of WIAT or WRAT. The editions of WIAT correlate well with each other (Burns, 2010; Johnson, 2011), many of the WRAT-3 items are also included in WRAT-4 (Dell et al., 2008), and both instruments are well-established tests of academic achievement. In addition, these instruments use the same type of scaled scores. Achievement scores, irrespective of instrument, were, thus, included in the analysis in the respective domain of academic achievement. Meta-analysis with grades could not be conducted, as grades were only used in two samples.

Full Scale IQ and academic achievement

As regards ADHD, a total of two studies (Carrasco et al., 2022; Mayes & Calhoun, 2007), which in turn included in total six samples (n = 1,112), were entered in meta-analyses focusing on the FSIQ. Three sets of outcome measures were possible to group for the meta-analysis, math (K = 6), word reading (K = 6), and written language (K = 6), see Figure 2. Regarding mathematics, the overall weighted correlation was .48 (95% CI: .21-.68). The heterogeneity value I^2 was 92%, indicating large heterogeneity. We note that the correlations reported by Mayes and Calhoun (2007) were larger than those of Carrasco et al. (2022). As regards word reading, the overall weighted correlation was .43 (95% CI: .27-.57). The heterogeneity value I^2 was 75%, indicating large heterogeneity. Again, the correlations reported by Mayes and Calhoun (2007) were larger than those of Carrasco et al. (2022). Regarding written language, the overall weighted correlation was .38 (95% CI: .24-.50). The heterogeneity value I^2 was 65%, indicating large heterogeneity. Due to the small number of samples, we did not perform a formal moderating analysis of the results presented above. There were too few studies on ASD to conduct a meta-analysis on this subgroup.

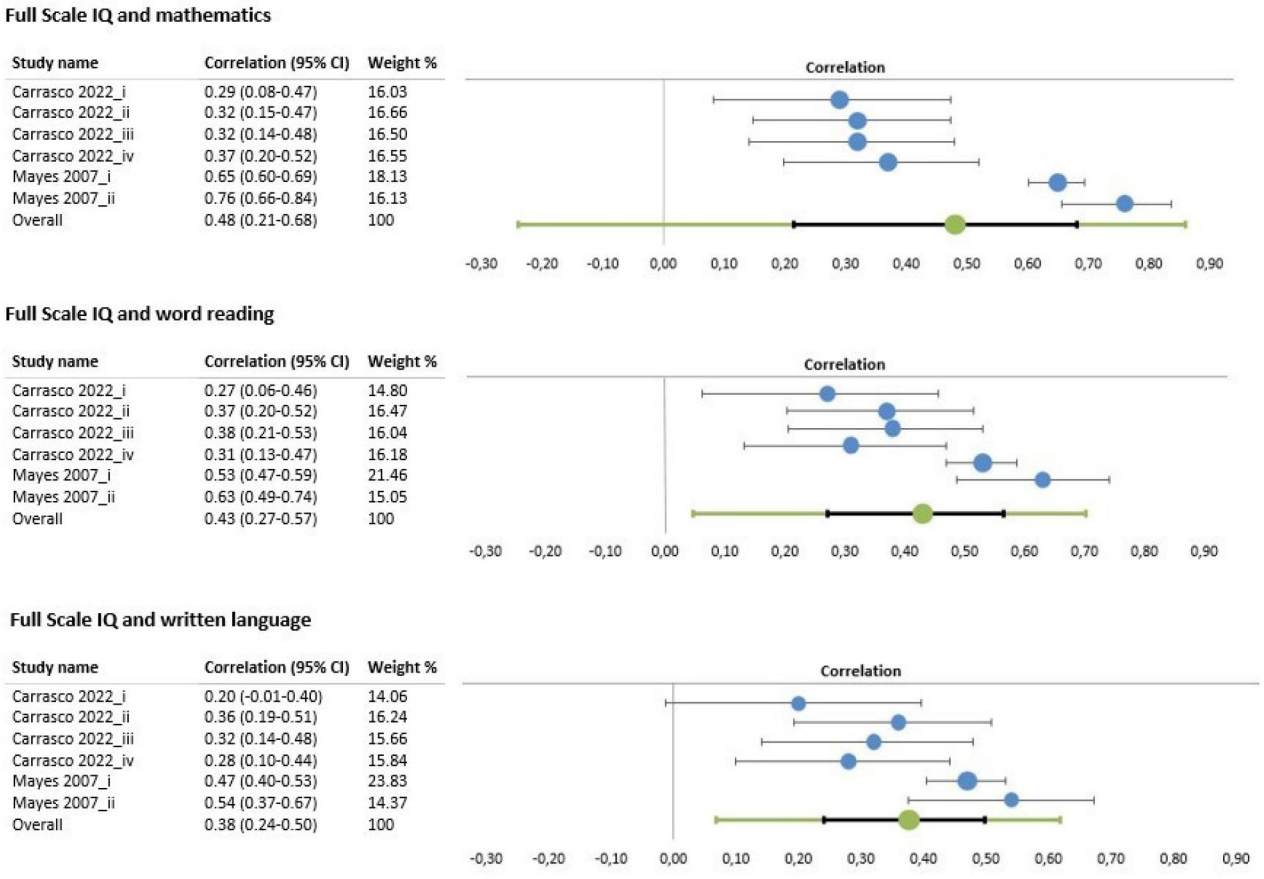


Figure 2. Forest plots of correlations between intellectual functioning and domains of achievement.
Note The horizontal bars represent 95% CIs. The green bars represent the prediction interval of the overall correlations.

Processing speed and academic achievement

As regards ADHD, the same two studies as above (Carrasco et al., 2022; Mayes & Calhoun, 2007) including six samples (n = 1,112) were entered in meta-analyses focusing on PSI. Three sets of outcome measures were possible to group for the meta-analysis, math (K = 6), word reading (K = 6), and written language (K = 6), see Figure 3. Regarding mathematics, the overall weighted correlation was .40 (95% CI: .33-.46). The heterogeneity value I^2 was 4%, not indicating heterogeneity. As regards word reading, the overall weighted correlation was .29 (95% CI: .14-.42). The heterogeneity value I^2 was 60%, indicating heterogeneity. The correlations reported by Carrasco et al. (2022) were in the small range, whereas those of Mayes and Calhoun (2007) were moderately sized. Regarding written language, the overall weighted correlation with processing speed was .35 (95% CI: .26-.43). The heterogeneity value I^2 was 32%. Again, there were too few studies on ASD to conduct a meta-analysis on this subgroup.

Publication bias

Due to the small number of samples included in the meta-analysis and presence of large heterogeneity in part of the analyses, considerable caution is required when interpreting results from analysis on publication bias. For transparency, funnel plots including results of trim-and-fill procedure were generated for the studies included in the meta-analyses and are provided in the supplementary material together with results of the trim-and-fill procedure (S2-S7).

Discussion

This systematic review included 12 studies that examined the associations between Wechsler scales and academic achievement and/or grades in children with ADHD or ASD. A total of six samples with ADHD from two of those studies were included in a meta-analysis. Of the 12 studies, three had low risk of bias and nine had medium risk of bias, suggesting that the results might be liable to bias, however, probably not enough to invalidate the results. The main finding of the systematic review is that FSIQ is associated with achievement in ADHD and ASD, and influences all studied domains (reading, written language, and mathematics). The abilities measured with WISC indexes (verbal reasoning and conceptualization, perceptual reasoning, working memory and processing speed) are all implicated in achievement, but the magnitude of impact of WISC indexes vary among studies. The number of studies examining WISC results and grades in ADHD is limited, and none of the included studies examined this relationship in ASD. In ADHD, higher VCI seems to increase the likelihood of passing grades. In the meta-analyses, FSIQ and PSI predicted academic achievement in reading, written language, and mathematics to moderate degree in ADHD.

The included ADHD studies demonstrated correlations between IQ/FSIQ and academic achievement (Carrasco et al., 2022; Mayes & Calhoun, 2007; Zendarski et al., 2017), and that FSIQ/IQ was, in relation to the other WISC indexes, the strongest predictor of academic achievement (Mayes & Calhoun, 2007; Zendarski et al., 2017). One study

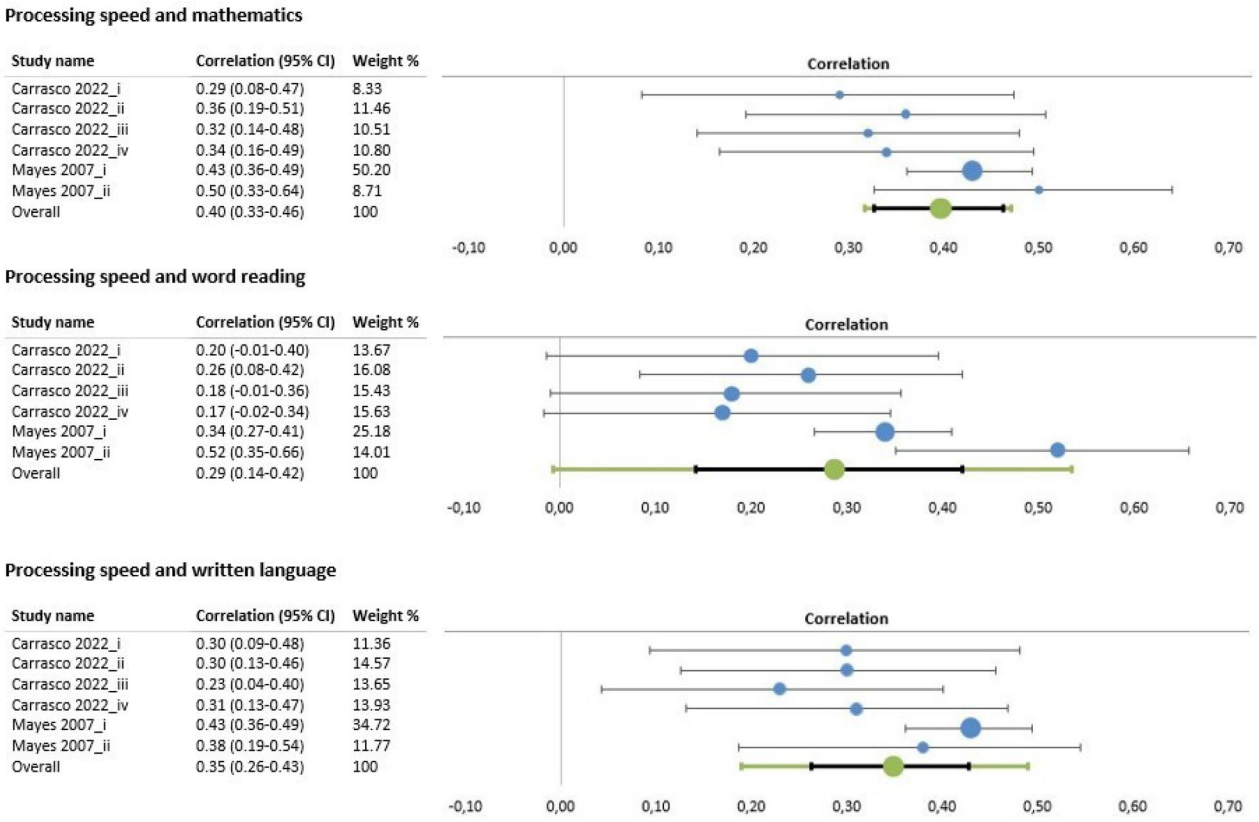


Figure 3. Forest plots of correlations between processing speed and domains of achievement.
Note The horizontal bars represent 95% CIs. The green bars represent the prediction interval of the overall correlations.

(Krane & Tannock, 2001) focused exclusively on working memory (WISC-III FDI) and found correlations between FDI with reading and with mathematics. The impact of working memory on achievement has also been highlighted in other studies (Calub et al., 2019; Carrasco et al., 2022; Mayes & Calhoun, 2007), and working memory was found to mediate academic achievement deficits in ADHD (Calub et al., 2019). However, Zendarski et al. (2017) did not find working memory to be independently associated with achievement (NAPLAN scores) when other explanatory variables (such as FSIQ, parent education, inattention, conduct problems) were included in the regression model. Instead, they found that FSIQ explained the greatest proportion of variance in NAPLAN scores. The authors discussed that this could be either due to working memory measure not being sensitive enough to capture working memory deficits or due to different academic achievement tests involving different aspects of working memory.

As regards other indexes, VCI was also implicated in academic achievement (Calub et al., 2019). Vocabulary subtest was a predictor of the variance in reading (Carrasco et al., 2022), spelling (Alloway & Stein, 2014; Carrasco et al., 2022), and arithmetic (Alloway & Stein, 2014). In addition, PRI correlated with scores of achievement (Alloway & Stein, 2014; Mayes & Calhoun, 2007), but did not make a significant contribution in regression analysis including other index scores. The contribution of PSI in academic achievement was limited in Carrasco et al. (2022), and both VCI and WMI correlated higher with achievement than PSI in Mayes and Calhoun (2007). Furthermore, children with ADHD and below average scores in WMI/PSI, had lower scores in reading fluency and math fluency compared to children with ADHD and high average WISC-IV scores (Thaler et al., 2013). In sum, it seems that working memory, which is associated with academic achievement in typically developing children (Caemmerer et al., 2018; Cortés Pascual et al., 2019), is also associated with academic achievement in ADHD. This result is consistent with findings of recent systematic reviews on cognition and literacy in ADHD (McDougal et al., 2022), and cognition and mathematics (Kanevski et al., 2022). As expected from current literature (Caemmerer et al., 2018), verbal comprehension abilities including vocabulary are important for academic achievement in ADHD. Lower processing speed has been associated with academic difficulties (Cook et al., 2018), which is evident in some of the included studies, but the magnitude of the impact of processing speed is less clear in the narrative review.

Though numerous studies have examined IQ, WISC profiles, and academic achievement in ASD, only a few have focused on the relationship between Wechsler intelligence scales and academic achievement. In the Mayes and Calhoun (2003) groups academic achievement scores were consistent with IQ in most domains, and FSIQ was the best overall single predictor of reading, math, and writing achievement in Mayes and Calhoun (2008). All WISC indexes correlated with achievement and the highest correlations on the index level were WMI for reading, PRI for math, and PSI for written expression. Furthermore, in an intellectually gifted ASD group, Assouline et al. (2012) found that both WMI and PSI correlated significantly with academic achievement and predicted a significant proportion of the variance in reading. WMI predicted variance in written language, and PSI together with other factors predicted a significant proportion of the variance in mathematics. Notably, VCI did not predict any scores of academic achievement, and PRI only predicted oral language. Generally, FSIQ, WMI, and PSI emerge as significant predictors of academic achievement in ASD, which is in line with the knowledge on IQ, working memory, processing speed, and academic achievement in other groups (Neisser et al., 1996; Titz & Karbach, 2014; Wechsler, 2003a). In addition, PRI correlated with math, reading, and writing achievement in Mayes and Calhoun (2008), but only with oral language in Assouline et al. (2012). This shows that perceptual reasoning is clearly implicated in academic achievement, but due to the limited number of studies, it is unclear if the differences between studies are related to the level of intellectual functioning of the participants, use of different instruments or other factors.

As regards the studies including grades as outcome, Ek et al. (2011) found moderate correlations between FSIQ, VIQ, VCI and PSI, and overall grade at grade 9 in ADHD. Tallberg et al. (2021) found that higher VCI at baseline was associated with having a majority of passing grades at follow-up. Lower VCI is implicated in poor academic outcomes in ADHD. The study did not examine the relationship between FSIQ and grades. Taken together, the studies underline the importance of verbal reasoning, comprehension, and conceptualization, general intellectual functioning and processing speed, for school grades. The findings are in line with the well-established predictive importance of intellectual functioning for grades (Roth et al., 2015). Future longitudinal studies could further examine the relationship between IQ and academic achievement in NDDs through exploring possible bidirectionality in these relations.

In the meta-analysis, including six ADHD samples, the overall weighted correlations between FSIQ and the three aspects of academic achievement (reading, written language, and mathematics) were moderate in size. Similarly, the overall weighted correlations between PSI and written language, and PSI and math, were moderate, and the correlation between PSI and reading was almost moderate. Notably, the correlations reported by Mayes and Calhoun (2007) are in some cases larger than those of Carrasco et al. (2022). This could be due to differences in sample characteristics such as mean FSIQ, which was slightly higher in Mayes and Calhoun (2007), comorbidities, and ADHD symptom severity. These factors are also mentioned by Carrasco et al. (2022) as regards differences between the samples included in their study. Furthermore, the overall weighted correlations found in the meta-analysis are lower than the correlations between WISC-IV and WIAT-II reported in the WISC-IV technical manual from a general population sample (Wechsler, 2003b). This also the case for the correlations

reported in Carrasco et al. (2022), while the correlations reported in Mayes and Calhoun (2007) are more in line with those reported in the manual, except for the correlations between FSIQ and written language, and between PSI and written language that are lower in Mayes and Calhoun (2007). These findings suggest that other characteristics related to ADHD or comorbidities may affect the strength of the relationship between intellectual functioning and academic achievement in ADHD. The included samples differed in terms of comorbidity, ADHD subtypes, and mean FSIQ, which reflects the heterogeneous nature of ADHD and ASD.

Furthermore, as reflected in the magnitude of correlations between cognitive domains and achievement, other factors, though not as strong as IQ, also need to be considered when studying and improving academic achievement in ADHD and/or ASD. Motivation, for example, also plays a significant role in academic achievement (Lavrijsen et al., 2022; Smith et al., 2020). Besides working memory, other executive functions (inhibitory control, cognitive flexibility) are related to academic achievement (Zelazo & Carlson, 2020). Executive function deficits affect academic achievement in ADHD (Biederman et al., 2004) and ASD (St. John et al., 2018), and cognitive training may improve academic outcomes (Veloso et al., 2019). In addition, ordinary classroom environment normally includes a variety of sensory stimuli and distractions, and poses demands on flexibility and other executive functions, which may be challenging for many children with ADHD and/or ASD (Ashburner et al., 2010; Kos et al., 2006). This stresses the importance of educational accommodations and interventions, such as close family-school partnership, behavioral classroom interventions, and individualized instructional supports that are recommended in ADHD (Wolraich et al., 2019), and visual supports and self-monitoring strategies that have been found effective in ASD (Watkins et al., 2019).

The evidence presented in this review has several limitations. Though the academic achievement instruments used in the studies are well established, there is some variation in selected versions and measures, limiting the number of samples included in the meta-analysis, which is a clear limitation. Another limitation for making general conclusions relates to the limited number of studies with participants with ASD and the relatively small – albeit not uncommon in clinical research- samples, requiring some caution when generalizing results, particularly on specific WISC indexes and domains of academic achievement. Lastly, factors such as socioeconomic status and school supports were not always reported in the studies.

Conclusion

Wechsler intelligence scales are frequently used for assessing intellectual functioning in ADHD (Thaler et al., 2013) and ASD (Linnenbank et al., 2022). The results of the present meta-analysis, which was also supported in the narrative synthesis, show that WISC FSIQ and PSI have bearing in predicting academic achievement in reading, written language, and mathematics to moderate degree in ADHD. This association is less studied in ASD, but results of existing studies provide support for association between WISC scores and academic achievement. Wechsler scales are, thus, valuable in predicting academic achievement in ADHD and ASD, but other factors also should be considered when predicting academic achievement in these groups.

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